

# AR LEAKAGE AUDIT

Power BI Dashboard — Technical & Analytical Documentation

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4 Pages | 16 Visuals | 6 DAX Measures | 3 Synced Slicers

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Companion Document to: SQL Analysis Report  
(AR\_Audit\_Project\_Documentation.pdf)

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# 1. DASHBOARD PURPOSE & AUDIENCE

This Power BI dashboard operationalises the SQL findings from the AR Leakage Audit into an interactive, self-service tool. While the SQL report answers fixed business questions, this dashboard allows a Finance leader, Collections Manager, or Sales Head to slice the same ₹8.17 Crore invoice base by date, payment status, or customer segment in real time — without writing a single query.

| Page                       | Primary Audience            | Core Question Answered                                    |
|----------------------------|-----------------------------|-----------------------------------------------------------|
| Executive Overview         | CFO / Finance Director      | Are we healthy overall? Are we improving or declining?    |
| Aging & Risk               | Collections Manager         | Where exactly is the money stuck, and how urgent is it?   |
| Sales & Segment Performace | Sales Director / Controller | Who owns this risk — which reps, segments, dispute types? |
| Regional Intelligence      | Regional Finance Heads      | Which geography needs different credit policy?            |

## 2. DATA MODEL & DAX MEASURES

The dashboard runs on a single flat table (AR\_Clean\_Data, 500 rows) produced by the Excel Power Query pipeline. Six DAX measures power every KPI card, chart axis, and conditional format across all four pages.

| Measure                 | DAX Logic                                        | Used In                                      |
|-------------------------|--------------------------------------------------|----------------------------------------------|
| Total Billed            | SUM(Invoice_Amount)                              | P1 KPI card, all segment/region bars         |
| Total Collected         | SUM(Amount_Paid)                                 | P1 KPI card, regional scorecard              |
| Total Outstanding       | SUM(Outstanding_Amount)                          | P1, P2, P3, P4 — used in nearly every visual |
| Collection Efficiency % | DIVIDE(SUM(Amount_Paid), SUM(Invoice_Amount), 0) | P1 KPI card, trend line, P4 scorecard        |
| Avg DSO                 | AVERAGEX over Paid/Late_Paid rows of Days_To_Pay | P1 KPI card, P4 regional DSO line            |
| Overdue Count           | COUNTROWS filtered to Payment_Status = Overdue   | Supporting measure for cards and tooltips    |

**Key design decision:** Collection Efficiency % is stored as a true decimal ratio (0.7259) with Percentage number-formatting applied at display level — not pre-multiplied by 100 in the DAX itself. This is best practice: it keeps the underlying value mathematically correct for any further calculations (e.g. variance analysis) while still rendering as **72.59%** on every card and axis.

### 3. HOW TO READ THIS DASHBOARD

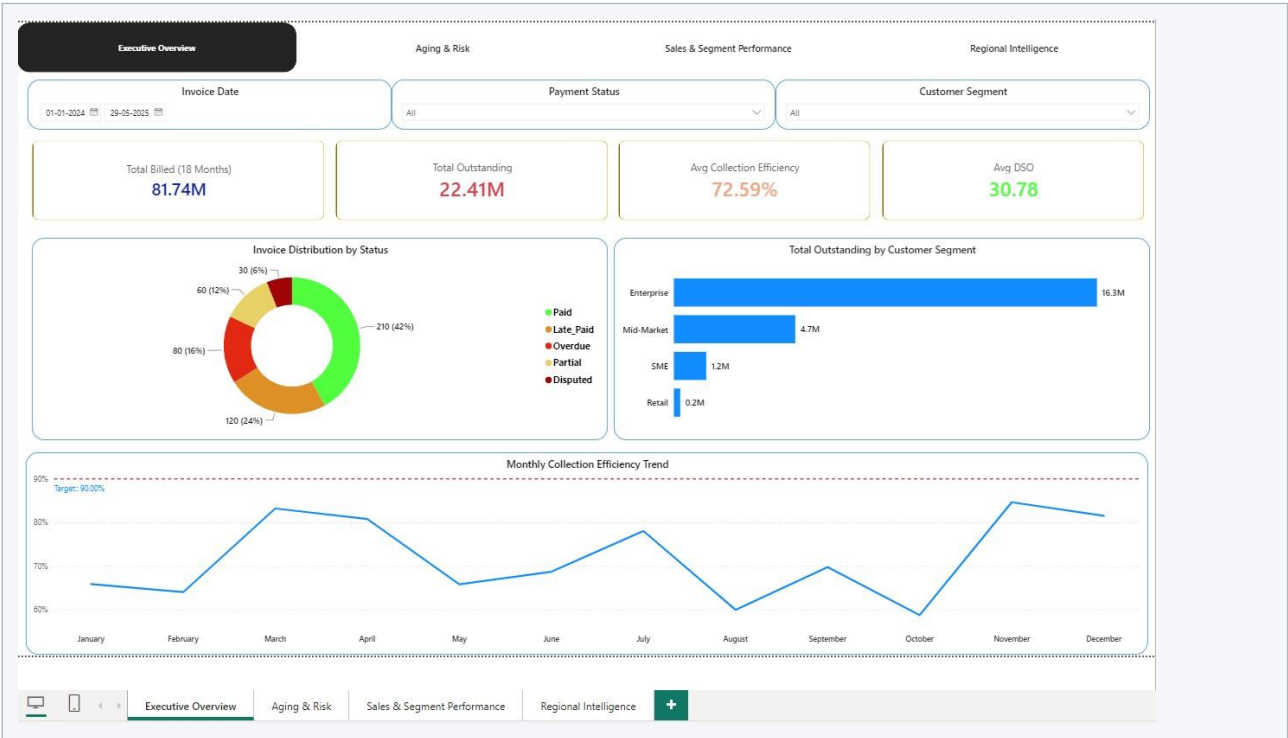
Every page shares the same three slicers at the top — Invoice Date, Payment Status, and Customer Segment — so a user can filter once and watch all four pages update in sync, thanks to Power BI's page navigation buttons retaining slicer state.

| Color                       | Meaning Across Dashboard                                  |
|-----------------------------|-----------------------------------------------------------|
| Dark Red                    | Critical — 90+ days overdue, highest financial risk       |
| Red / Orange                | High priority — disputed or significantly overdue         |
| Green                       | Healthy — paid, low-risk, or within 0-30 day window       |
| Grey                        | Neutral / no action needed (e.g. None priority bucket)    |
| Dark Blue gradient (Matrix) | Heatmap intensity — darker = higher invoice concentration |

**Navigation:** Four buttons (Executive Overview, Aging & Risk, Sales & Segment Performance, Regional Intelligence) appear at the top of every page. The active page is highlighted in black. Clicking any button jumps instantly to that page while preserving all active slicer selections.

## 4. PAGE 1 — EXECUTIVE OVERVIEW

The CFO's first stop. Four KPI cards establish the headline numbers, followed by a status breakdown, a segment risk bar, and a 17-month efficiency trend against a 90% target line.



### Visual 1 — KPI Cards (×4)

Type: Card | Fields: Total Billed, Total Outstanding, Avg Collection Efficiency, Avg DSO

Total Billed stands at ₹81.74M (₹8.17 Cr) across 18 months. Total Outstanding is ₹22.41M (₹2.24 Cr) — 27.4% of everything billed has not been collected. Average Collection Efficiency of 72.59% sits 17+ points below the 90% industry benchmark. Average DSO of 30.78 days is marginally above the 30-day target, indicating the business is, on average, just at the edge of acceptable but with high variance underneath.

### Visual 2 — Invoice Distribution by Status

Type: Donut Chart | Fields: Legend: Payment\_Status | Values: Count of Invoice\_ID

42% of invoices (210) are cleanly Paid and 24% (120) are Late\_Paid — together a 66% core that eventually settles. The remaining third splits across Overdue (16%, 80 invoices), Partial (12%, 60), and Disputed (6%, 30) — the three categories driving the entire ₹22.41M outstanding figure.

### Visual 3 — Total Outstanding by Customer Segment

Type: Bar Chart | Fields: Y-axis: Customer\_Segment | X-axis: Total Outstanding measure

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Enterprise customers alone account for ₹16.3M of the ₹22.41M total outstanding — roughly 73% of all locked cash sits with the segment that should, in theory, have the strongest credit discipline.

Mid-Market (₹4.7M), SME (₹1.2M) and Retail (₹0.2M) trail far behind in absolute terms.

### Visual 4 — Monthly Collection Efficiency Trend

Type: Line Chart + Constant Line | Fields: X-axis: Invoice\_Date (Month) | Y-axis: Collection Efficiency % | Constant line at 90%

The trend line never once touches the 90% target line across the full 17-month window. It peaks near 84-87% around March and November but dips below 65% in January, May, and August — a clear seasonal volatility pattern rather than steady improvement.

#### ■ Insight

Enterprise carries 73% of outstanding value on only 70 of 500 invoices (14% of volume) — this is a concentration problem, not a volume problem.

Separately, the efficiency dip in January/May/August (visible on the trend line) is a pattern worth testing against invoice issuance volume in the preceding month — Indian B2B billing often spikes at fiscal quarter-ends, and a surge in invoices raised in Q-end month could mechanically depress next month's efficiency ratio simply because more invoices are still within their 30-day window. This is a hypothesis the raw data supports investigating further, not yet a confirmed cause.

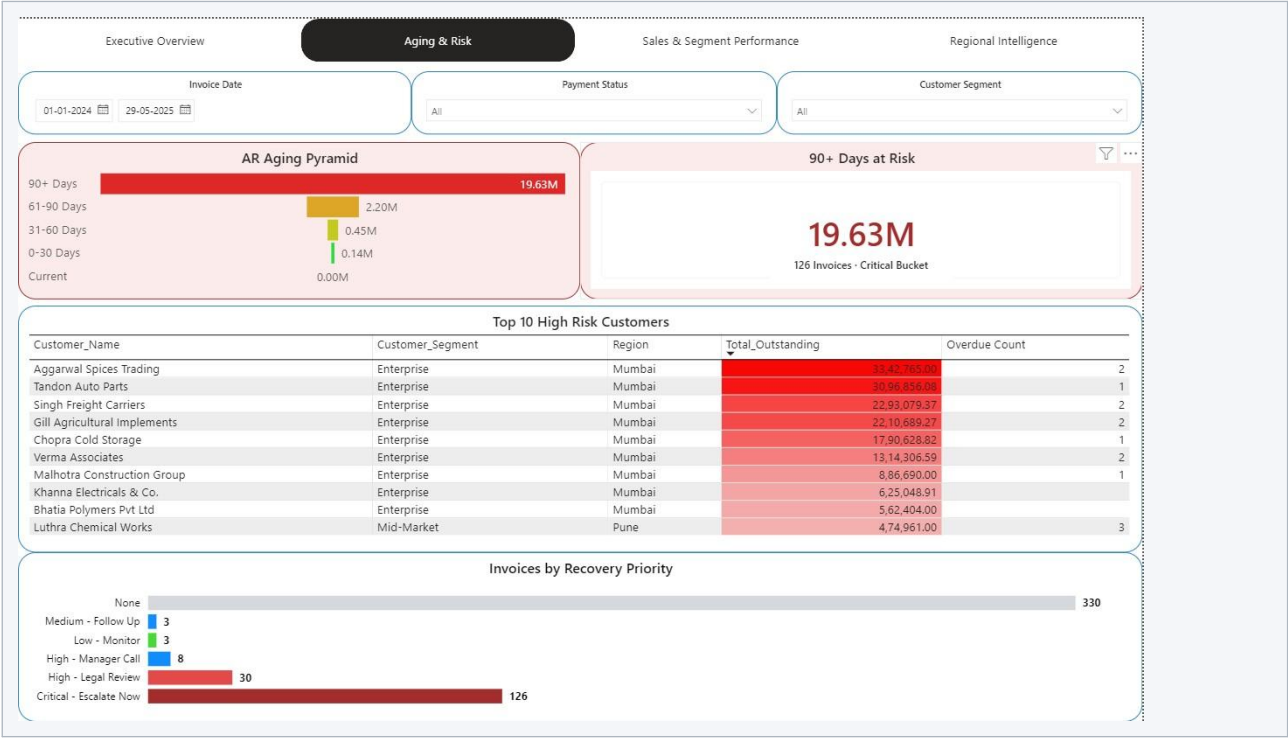
#### ■ Business Action

Recovering just the Top 10 accounts shown on Page 2 (₹1.66 Cr, which is 74% of all outstanding) would lift Collection Efficiency from 72.59% to approximately **92.9%** — a +20-point swing from fixing 10 relationships out of 80 customers. This single number should anchor the entire collections strategy: this is fundamentally a concentrated, not distributed, problem.

*Note on methodology: the Top 10 recovery calculation above assumes full collection of all 10 accounts' outstanding balances and holds Total Billed constant — it is a best-case ceiling, not a guaranteed forecast.*

## 5. PAGE 2 — AGING & RISK

The Collections Manager's working page. Combines an aging pyramid, a dedicated 90+ day risk card, a ranked customer table with in-cell heat bars, and a recovery priority breakdown. This page's figures changed materially after a Days\_Overdue calculation fix — see the revision note below before the visual breakdown.



### Visual 1 — AR Aging Pyramid

Type: Funnel Chart | Fields: Group: Aging\_Bucket | Values: Total Outstanding

90+ Days now holds ₹19.63M (87.6% of all outstanding) versus ₹0.14M in 0-30 Days. The 31-60 Day (₹0.45M) and 61-90 Day (₹2.20M) buckets.

### Visual 2 — 90+ Days At Risk

Type: Card with visual-level filter | Fields: Total Outstanding measure | Filter: Aging\_Bucket = 90+ Days

₹19.63M sits across 126 invoices in the critical bucket on this card. Note: this 126 count reflects the Recovery\_Priority field's 'Critical' category specifically, which by design excludes Disputed invoices (routed instead to 'High - Legal Review' regardless of age — see Visual 4). A parallel SQL query against the raw Aging\_Bucket field, which makes no such carve-out, counts 154 invoices in the 90+ Day



window. Both numbers are correct; they answer slightly different questions (recovery routing vs. raw aging), and the distinction matters when briefing the CFO on headcount-of-invoices versus rupee exposure.

### Visual 3 — Top 10 High Risk Customers

Type: Table with in-cell heat bars | Fields: Customer\_Name, Customer\_Segment, Region, Total\_Outstanding, Overdue\_Count

Unaffected by the Days\_Overdue fix — this table is built from Total\_Outstanding, not aging. All top 10 are Mumbai-based, and 9 of 10 are Enterprise segment. Aggarwal Spices Trading (₹33.4L) and Tandon Auto Parts (₹31.0L) top the list. The heat-bar shading makes the severity hierarchy immediately scannable without reading every number.

### Visual 4 — Invoices by Recovery Priority

Type: Bar Chart | Fields: Y-axis: Recovery\_Priority | X-axis: Count of Invoice\_ID

None (330), Critical-Escalate, Now (126), High-Legal Review (30, all Disputed invoices by design), High-Manager Call (8), Medium-Follow Up (3), and Low-Monitor (3).

#### ■ Insight

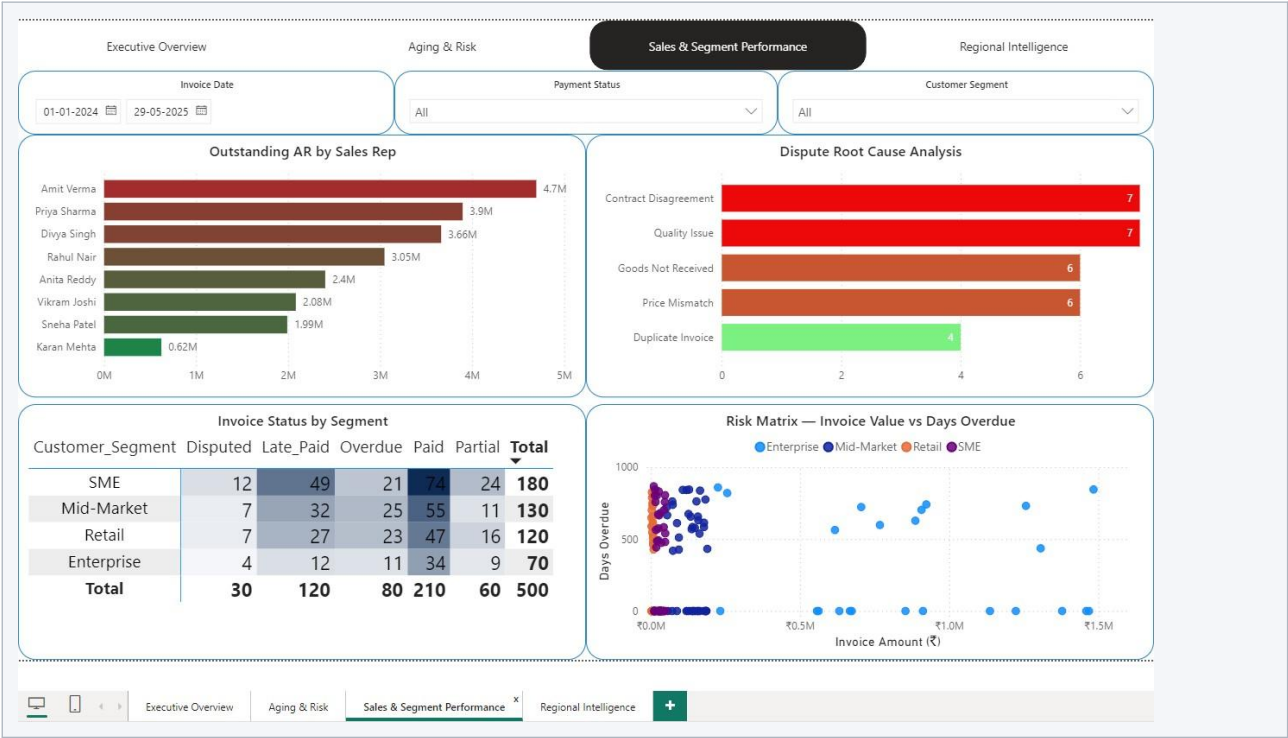
The business does have a working — if thin — mid-stage collections pathway: 11 invoices now genuinely sit in the 31-90 day range (8 High-Manager Call + 3 Medium-Follow Up), proving invoices can be caught before they become critical. The real problem is scale: 126-154 invoices (depending on which field is used) are already in the 90+ Day zone, dwarfing the mid-stage pipeline.

#### ■ Business Action

Strengthen the mid-stage process that the corrected data shows does exist, rather than building one from scratch. Prioritise getting the 126+ critical invoices into a structured write-off/legal review process (Section 9), since routine follow-up at this stage has a low success rate.

## 6. PAGE 3 — SALES & SEGMENT PERFORMANCE

Accountability page connecting AR outcomes back to the people and processes that create them — sales reps, dispute root causes, segment-level status mix, and an invoice-level risk scatter.



### Visual 1 — Outstanding AR by Sales Rep

Type: Bar Chart with color gradient | Fields: Y-axis: Sales\_Rep | X-axis: Total Outstanding | Color: Gradient (green→red) on value

Amit Verma (₹4.7M) and Priya Sharma (₹3.9M) carry the heaviest overdue portfolios — together over ₹8.6M, nearly 39% of total outstanding from just 2 of 8 reps. Karan Mehta, by contrast, holds only ₹0.62M — the gradient color makes this contrast immediately visible without reading exact figures.

### Visual 2 — Dispute Root Cause Analysis

Type: Bar Chart | Fields: Y-axis: Dispute\_Reason | X-axis: Count of Invoice\_ID | Filter: Dispute\_Flag = Yes

Contract Disagreement and Quality Issue are tied as the top dispute drivers (7 cases each), followed by Goods Not Received and Price Mismatch (6 each). These four reasons account for 26 of 30 total disputes — concentrated, addressable root causes rather than random one-offs.

#### Visual 3 — Invoice Status by Segment (Matrix)

Type: Matrix with heatmap formatting | Fields: Rows: Customer\_Segment | Columns: Payment\_Status | Values: Count, color-scaled

SME has the highest invoice volume (180) and the darkest 'Paid' cell (74), confirming small accounts transact more frequently. Mid-Market's Paid count (55) is its strongest cell. Enterprise has the lowest volume (70) but, as Page 1 showed, the highest outstanding value — a classic low-volume, high-value risk profile.

#### Visual 4 — Risk Matrix — Invoice Value vs Days Overdue

Type: Scatter Chart | Fields: X-axis: Invoice\_Amount | Y-axis: Days\_Overdue | Legend: Customer\_Segment | Details: Invoice\_ID

Enterprise dots (light blue) spread across the highest invoice values (₹0.5M–₹1.5M+) with overdue days ranging 400-850 — the most dangerous combination on the chart. SME and Retail dots cluster tightly at low invoice values but still show overdue ranges up to 1000 days, proving small-ticket invoices can be neglected just as long as large ones.

##### ■ Insight

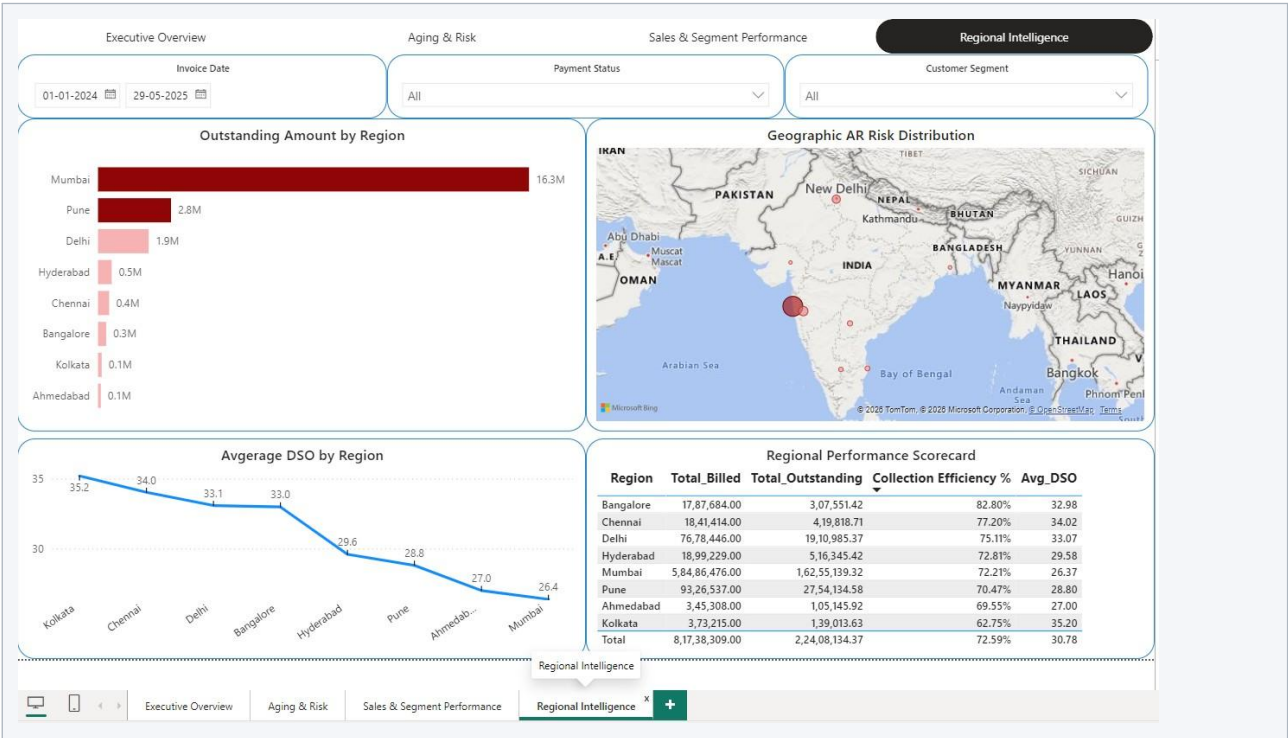
AR risk is not evenly distributed across the sales team — two reps account for nearly 40% of overdue value. Disputes cluster around four specific, fixable root causes rather than being random noise.

##### ■ Business Action

Make Amit Verma and Priya Sharma's accounts the subject of a joint Sales-Finance review this month. Fix Contract Disagreement and Quality Issue disputes at the source by mandating clearer Scope-of-Work documentation before invoicing.

## 7. PAGE 4 — REGIONAL INTELLIGENCE

Geography-based view for regional finance heads — outstanding by region, a map visualisation, a DSO-by-region trend, and a full regional scorecard table.



### Visual 1 — Outstanding Amount by Region

Type: Bar Chart | Fields: Y-axis: Region | X-axis: Total Outstanding measure

Mumbai dominates with ₹16.3M outstanding — more than 5x the next closest region (Pune at ₹2.8M). This mirrors Page 1’s segment finding: Mumbai is where most Enterprise customers are based, so the regional and segment risk stories are really the same underlying concentration.

### Visual 2 — Geographic AR Risk Distribution

Type: Map Visual | Fields: Location: Region | Bubble Size: Total Outstanding

The map bubble over Mumbai dwarfs every other Indian city shown, giving an instant geographic confirmation of where collections effort should concentrate physically — useful if planning in-person visits or assigning region-based collection agents.

### Visual 3 — Average DSO by Region

Type: Line Chart | Fields: X-axis: Region (sorted) | Y-axis: Avg DSO measure

Kolkata has the worst DSO at 35.2 days, while Mumbai shows the best DSO at 26.37 days — despite Mumbai also holding the highest absolute outstanding value on the chart to its left. This looks contradictory at first glance and is resolved explicitly below.

#### Visual 4 — Regional Performance Scorecard

Type: Table sorted by Collection Efficiency | Fields: Region, Total\_Billed, Total\_Outstanding, Collection Efficiency %, Avg\_DSO

Bangalore leads at 82.80% collection efficiency; Kolkata trails at 62.75% — a 20-point gap between best and worst region. Mumbai sits mid-table at 72.21%, almost exactly the company average of 72.59% — confirming Mumbai is large in absolute terms but not proportionally worse-behaved than the portfolio as a whole.

#### Resolving the Mumbai Paradox: Why Best DSO and Worst Outstanding Coexist

The Avg DSO measure is deliberately built as: `AVERAGEX(FILTER(Payment_Status IN {Paid, Late_Paid}), Days_To_Pay)`. By construction, this formula only ever looks at invoices that **did** eventually get paid — it structurally excludes every Overdue, Partial, and Disputed invoice. DSO and Outstanding Amount are therefore not measuring the same population of invoices; they are two different lenses on the same region.

| Metric                                   | What It Measures                                | Mumbai Value                                                        |
|------------------------------------------|-------------------------------------------------|---------------------------------------------------------------------|
| Avg DSO (26.37 days)                     | Speed of the invoices that DID get paid         | Best in the dataset — Mumbai's healthy invoices pay fast            |
| Total Outstanding (₹16.3M)               | Total stuck in invoices that have NOT been paid | Worst in absolute terms — driven by ~9-10 chronic accounts          |
| Outstanding as % of Mumbai's own billing | 27.87% of everything billed in Mumbai is unpaid | In line with company average (27.4%) — not actually an outlier rate |

**Conclusion:** Mumbai does not have a payment-speed problem — it has a concentration problem, identical to the Enterprise-segment finding on Page 1 (this is the same root cause appearing twice from two different angles, which strengthens confidence that it is real and not a fluke of one chart's filtering). Kolkata and Ahmedabad, by contrast, show weak metrics on every cut — they have an actual behavioural payment problem across their customer base, not a concentration issue.

#### ■ Insight

Two regions need attention for entirely different reasons: Mumbai's issue is 9-10 specific accounts (a collections task), while Kolkata/Ahmedabad's issue is portfolio-wide behaviour (a credit policy task). Conflating them into one 'regional risk' response would misallocate effort.

#### ■ Business Action

Mumbai: assign the Top 10 table (Page 2) to named account owners — no regional policy change needed. Kolkata & Ahmedabad: tighten terms for ALL customers (e.g. shorten credit period from 30 to 15 days, or require partial advance) since the weakness is structural, not isolated to a few accounts.

## 8. DATA LIMITATIONS & HONEST CAVEATS

A strong analytics portfolio acknowledges what the data does NOT show, not just what it does. The two limitations remain genuinely unresolved.

### Synthetic, not live, data

All 500 invoices were constructed to realistically mirror AR patterns observed in real Finance/AR experience, but are not drawn from an actual company's transaction system. Numbers and trends are illustrative of the analytical method, not a real business's financial position.

### Map visual approximates city centroids

Power BI's built-in geocoding places each region at its city centre point; it does not reflect customer-specific addresses, so the bubble size is accurate but its exact pin location is a city-level estimate.

### Recovery\_Priority and raw Aging\_Bucket count Disputed invoices differently

Recovery\_Priority always routes Disputed invoices to 'High - Legal Review' regardless of how old they are, while the raw Aging\_Bucket field ages them normally. This means the '90+ Days at Risk' card (126 invoices, built on Recovery\_Priority) and a direct SQL count against Aging\_Bucket (154 invoices) will legitimately disagree — both are correct, but this dual-counting convention should be documented for anyone extending this model, to avoid it looking like a bug.

## 9. CROSS-PAGE SYNTHESIS & QUANTIFIED RECOMMENDATIONS

Reading all four pages together reveals one consistent story, confirmed independently three separate ways: by segment (Page 1), by named customer (Page 2), and by region (Page 4). **A small, identifiable concentration of Enterprise/Mumbai accounts — not the customer base broadly — is responsible for the majority of AR risk.** The fact that three unrelated cuts of the same 500 rows point to the same handful of accounts is what makes this finding trustworthy rather than coincidental.

### Scenario Modelling — What Moves the 72.59% Number

The 0-30 Day bucket holds only ₹0.14M, which sharpens the conclusion: there is no large 'easy win' bucket, and the Top 10 account strategy is the dominant lever by a wide margin.

| Scenario                                                  | Amount Recovered | New Outstanding | New Efficiency | Point Gain        |
|-----------------------------------------------------------|------------------|-----------------|----------------|-------------------|
| Current state                                             | —                | ₹22.41M         | 72.59%         | —                 |
| Recover 0-30 Day bucket only (corrected near-term target) | ₹0.14M           | ₹22.27M         | 72.76%         | +0.17 pts         |
| Recover 0-30 + 31-60 Day buckets combined                 | ₹0.59M           | ₹21.82M         | 73.31%         | +0.72 pts         |
| <b>Recover Top 10 accounts fully (Page 2 table)</b>       | <b>₹16.60M</b>   | <b>₹5.81M</b>   | <b>92.89%</b>  | <b>+20.31 pts</b> |

*All scenarios assume Total Billed (₹81.74M) stays constant and recovered amounts convert fully from Outstanding to Collected — these are best-case ceilings to set targets against, not forecasts of what will happen.*

The Top 10 scenario delivers roughly **28 times** the efficiency gain of recovering the entire 0-30 and 31-60 Day buckets combined (+20.31 pts vs. +0.72 pts), despite touching only 10 of 500 invoices (2% of volume). This is the single most important number in the entire analysis: **there is no meaningful 'quick win' bucket; collections bandwidth should go almost entirely to the Top 10 accounts.**

### The Five-Point Action Plan, in Priority Order:

#### 1. Assign named owners to the Top 10 accounts this week

Highest leverage action available: ₹1.66 Cr (74% of all outstanding) sits with 10 customers. This alone is worth +20.3 efficiency points — more than 28 times the combined near-term bucket, and the only intervention modelled above worth prioritising over the others.

## 2. Strengthen the existing 30-60-90 day escalation pathway

The data confirms 11 invoices genuinely sit in the 31-90 day range (8 High-Manager Call + 3 Medium-Follow Up) — a thin but real mid-stage pipeline. The priority is scaling this pathway's capture rate, not building one from scratch: most invoices still slip past it into the 126+ invoice, ₹19.63M critical bucket before being caught.

## 3. Differentiate regional policy by mechanism, not magnitude

Mumbai needs account-level collections (a people problem, 9-10 names). Kolkata and Ahmedabad need policy-level change — shorter credit terms for everyone — because their weakness shows up on every metric (DSO, efficiency, outstanding %), not just one.

## 4. Pair sales accountability with the four named dispute causes

Amit Verma and Priya Sharma hold 39% of overdue value between two reps; Contract Disagreement and Quality Issue cause the most disputes. Link commission to collected (not billed) revenue, and mandate Scope-of-Work sign-off before invoicing for Consulting engagements specifically.

## 5. Document the Recovery\_Priority / Aging\_Bucket dual-counting convention

The remaining follow-up item is documentation, not calculation: Recovery\_Priority and the raw Aging\_Bucket field now legitimately disagree on Disputed-invoice counts by design (126 vs. 154 in the 90+ bucket) — this convention should be written into the data dictionary before anyone else extends this model, so it isn't mistaken for a new bug.